Heat-engine efficiency

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

At the same kelvin temperature, sample B contains 3 times as many particles of the same ideal gas as A. Compare average energy per particle and total translational energy.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = 1 - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

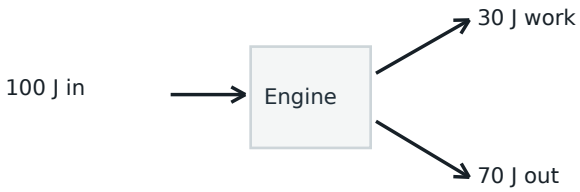
Worked reasoning: A heat engine takes 100 J from a hot source, does 25 J of work and rejects 75 J. Efficiency=25/100=25%.

Your turn: Draw three labeled energy arrows and check conservation.

Marzano check - circle: ① ② ③

03 Look and explain

An engine receives 100 J, produces 30 J of work and rejects 70 J. Check the energy balance and find efficiency.



Marzano check - circle: ① ② ③

04 Build the solution

A heat engine receives 200 J and produces 50 J useful work. Find efficiency and rejected heat.

Marzano check - circle: ① ② ③

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05 Use the idea

Why should a 95% efficient electric motor not be confused with a 95% heat engine converting heat from one source?

Marzano check - circle: 1 2 3

06 Choose a method

At fixed mass, speed becomes 1/4 of its original value. Compare kinetic energy and the energy-equivalent TNT amount.

Marzano check - circle: 1 2 3

07 Exit check

A device outputs 40 J work from 100 J heat. State efficiency and rejected heat.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

An engine claims 100 J input, 40 J useful work and 50 J rejected heat. Identify the missing accounting and give one possible explanation.

Marzano check - circle: 1 2 3